

Strictly Convex Targets in the $L_1(\mu, X)$ *CWO-S* Question

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Status

This packet gives a strengthened partial result for the natural question of Haller–Kuuseok–Poldvere [1]. The source paper proves that $L_1(\mu, X)$ has property *CWO-S* when X is weakly uniformly rotund (*wUR*), and asks whether *wUR* can be replaced by the formally weaker assumption that X itself has property *CWO-S*.

The result here proves the source theorem for every strictly convex Banach space X , with no Radon–Nikodym assumption on X^* . This is still not the full X has *CWO-S* problem, because there are *CWO-S* spaces that are not strictly convex, and it does not settle the injective tensor-product question.

Source Problem Evidence

The source question appears in the introduction of [1], page 3 of the arXiv PDF:

Then also the 1-sum $\bigoplus_{i=1}^{\infty} X_i$ has property *CWO-S*.

Finally, in Section 4, we prove Theorem 1.1.

A natural question arises in connection with Theorem 1.1: does this theorem remain true when X is assumed to be any Banach space with property *CWO-S* (note that strictly convex spaces—and thus, as a special case, *wUR* spaces—do have property *CWO-S*) or, moreover, does the injective tensor product of two Banach spaces with property *CWO-S* have property *CWO-S*?

2. AUXILIARY RESULTS FOR THE PROOF OF THEOREM 1.1

In [2, Theorem 5.5] it was proven that, for any non-zero σ -finite (countably additive non-negative) measure μ , the real space $L_1(\mu)$ has property *CWO-S*. However, the proof in [2] contains a gap: it is valid only for the case when, using the no-

Transcription

After proving the *wUR*-valued L_1 -theorem and countable ℓ_1 -sum stability for *CWO-S*, the authors ask whether the theorem remains true when X is assumed to be any Banach space with property *CWO-S*, and whether the injective tensor product of two *CWO-S* spaces must have property *CWO-S*.

Definitions

A Banach space X has property *CWO-S* if, for every finite convex combination $C = \sum_{k=1}^n \lambda_k U_k$ of relatively weakly open subsets of B_X , every point of $C \cap S_X$ is an interior point of C in the relative weak topology of B_X .

Proof Intuition

The source proof uses *wUR* to turn almost norm-attainment of local Bochner averages into weak closeness of the two local summands. Strict convexity gives a different mechanism. If $z = \lambda x + \nu y \in S_{L_1(\mu, X)}$, then equality in the L_1 -triangle inequality holds almost everywhere. In a strictly convex target this forces $x(\omega), y(\omega), z(\omega)$ to lie on the same positive ray almost everywhere. After partitioning the range of the unit direction $e(\omega) = z(\omega)/\|z(\omega)\|$ into small-diameter pieces, the local averages of x, y, z are norm-close in the only way needed by the decomposition argument. General weak functionals on $L_1(\mu, X)$ are handled by composing the finitely many functionals defining the weak neighbourhoods with the coordinate multipliers $f \mapsto \mathbf{1}_D f$.

Lemma 1 (Directional averaging). *Let X be a Banach space, let D be measurable, and suppose*

$$x = re, \quad y = se, \quad z = qe$$

on D , where $e : D \rightarrow S_X$, $r, s, q \geq 0$, $q = \lambda r + \nu s$, and $\lambda, \nu > 0$, $\lambda + \nu = 1$. Put

$$\alpha = \int_D r \, d\mu, \quad \beta = \int_D s \, d\mu, \quad \gamma = \int_D q \, d\mu.$$

If $\gamma > 0$ and $\text{diam } e(D) \leq \delta$, then

$$\left\| \frac{\alpha}{\gamma} z - x \right\|_{L_1(D)} \leq 2\delta\alpha, \quad \left\| \frac{\beta}{\gamma} z - y \right\|_{L_1(D)} \leq 2\delta\beta.$$

Proof. Choose $e_0 \in e(D)$. Since

$$\int_D \left(\frac{\alpha}{\gamma} q - r \right) d\mu = 0,$$

we have

$$\frac{\alpha}{\gamma} z - x = \left(\frac{\alpha}{\gamma} q - r \right) (e - e_0).$$

Also $q \geq \lambda r$, so $\alpha/\gamma \leq 1/\lambda$, and

$$\int_D \left| \frac{\alpha}{\gamma} q - r \right| d\mu \leq \frac{\alpha}{\gamma} \int_D q \, d\mu + \int_D r \, d\mu = 2\alpha.$$

The first estimate follows. The second is identical. □

Theorem 1. *Let μ be a non-zero σ -finite countably additive non-negative measure and let X be a strictly convex Banach space. Then*

$$L_1(\mu, X)$$

has property CWO-S.

Proof. By the countable ℓ_1 -sum stability theorem of [1, Proposition 1.2], it is enough to prove the result when μ is finite. Indeed, a σ -finite measure space decomposes into countably many finite-measure pieces, and $L_1(\mu, X)$ is the corresponding countable ℓ_1 -sum.

Set $Z = L_1(\mu, X)$, and let U, V be relatively weakly open subsets of B_Z . Let $x \in U$, $y \in V$, and $\lambda, \nu > 0$, $\lambda + \nu = 1$, satisfy

$$z = \lambda x + \nu y \in S_Z.$$

By the reduction lemma of [1, Lemma 2.1], we may assume that x and y have the same support up to null sets: wherever one is non-zero, so is the other. It suffices to find a relative weak neighbourhood W of z in B_Z such that $W \subset \lambda U + \nu V$.

Choose a finite set $F \subset S_{Z^*}$ and $\eta > 0$ so small that

$$U_0 = \{u \in B_Z : |f(u - x)| < \eta \text{ for every } f \in F\} \subset U$$

and

$$V_0 = \{v \in B_Z : |f(v - y)| < \eta \text{ for every } f \in F\} \subset V.$$

All constants below are chosen sufficiently small relative to η , λ , ν , and F .

Since $\|z\| = \lambda\|x\| + \nu\|y\| = 1$, equality holds in the pointwise triangle inequality almost everywhere. Strict convexity and the common support assumption imply that there are measurable functions

$$e : \Omega \rightarrow S_X, \quad r, s, q : \Omega \rightarrow [0, \infty)$$

such that

$$x = re, \quad y = se, \quad z = qe, \quad q = \lambda r + \nu s$$

almost everywhere on $\{z \neq 0\}$, and all four functions vanish off that set.

Choose $\beta > 0$ and a measurable set $G \subset \{z \neq 0\}$ so that

$$r/q \geq \beta, \quad s/q \geq \beta \quad \text{on } G,$$

and

$$\int_{\Omega \setminus G} (r + s + q) d\mu$$

is as small as needed. This is possible because r/q and s/q are positive almost everywhere on $\{z \neq 0\}$. Since e is strongly measurable, partition a further large subset of G into finitely many measurable sets $D_2, \dots, D_n$ on which $\text{diam } e(D_i) \leq \delta$, leaving a bad set D_1 with $\int_{D_1} (r + s + q) d\mu$ as small as needed. Put also $D_0 = \Omega \setminus (D_1 \cup \dots \cup D_n)$, which is null for z . We choose the partition so that

$$\Gamma := \sum_{i=2}^n \gamma_i > 0,$$

where

$$\alpha_i = \int_{D_i} r d\mu, \quad \beta_i = \int_{D_i} s d\mu, \quad \gamma_i = \int_{D_i} q d\mu.$$

On the good pieces $i \geq 2$, one has

$$\alpha_i/\gamma_i \geq \beta, \quad \beta_i/\gamma_i \geq \beta.$$

Using [1, Lemma 2.2(b)], choose a relative weak neighbourhood W_{mass} of z in B_Z such that, for $w \in W_{\text{mass}}$, the quantities

$$\eta_i(w) = \|\mathbf{1}_{D_i} w\| - \gamma_i, \quad i = 0, \dots, n,$$

satisfy $\sum_{i=0}^n |\eta_i(w)| < d$, and also $\|\mathbf{1}_{D_1} w\|$ is as small as needed. Here $d > 0$ is chosen so small that

$$\frac{2d}{\Gamma} \max\{1/\lambda, 1/\nu\} < \beta/2.$$

For each $f \in F$ and each $i = 1, \dots, n$, the map

$$w \mapsto f(\mathbf{1}_{D_i} w)$$

is a continuous linear functional on Z . Intersect W_{mass} with conditions making every

$$|f(\mathbf{1}_{D_i}(w - z))|$$

smaller than a chosen tiny constant. Call the resulting relative weak neighbourhood W .

Fix $w \in W$, and write $w_i = \mathbf{1}_{D_i} w$. For i with $\gamma_i = 0$, put $u_i = v_i = w_i$. For $i = 1$ with $\gamma_1 > 0$, put

$$u_1 = (\alpha_1/\gamma_1)w_1, \quad v_1 = (\beta_1/\gamma_1)w_1.$$

For the good pieces $i = 2, \dots, n$, define

$$\kappa = \sum_{\gamma_i > 0} \frac{\beta_i - \alpha_i}{\gamma_i} \eta_i(w), \quad \rho = \sum_{i=2}^n \|w_i\|.$$

The choice of d gives

$$|\kappa|/\rho < \beta/2.$$

Hence the following coefficients are non-negative:

$$\alpha_i/\gamma_i + \nu\kappa/\rho, \quad \beta_i/\gamma_i - \lambda\kappa/\rho \quad (i = 2, \dots, n).$$

Set

$$u_i = (\alpha_i/\gamma_i + \nu\kappa/\rho) w_i, \quad v_i = (\beta_i/\gamma_i - \lambda\kappa/\rho) w_i \quad (i = 2, \dots, n),$$

and finally $u = \sum_i u_i$, $v = \sum_i v_i$. Then $\lambda u + \nu v = w$.

The usual ℓ_1 -bookkeeping, identical to the computation in the proof of [1, Theorem 1.1], gives $\|u\| \leq 1$ and $\|v\| \leq 1$. Indeed, after summing over all pieces with $\gamma_i > 0$, the term containing κ changes

$$\sum_i (\alpha_i/\gamma_i) \|w_i\|$$

into

$$\sum_i \alpha_i + \sum_i \eta_i(w),$$

and the remaining $\gamma_i = 0$ pieces satisfy $x = y = z = 0$. Since $\sum_i \eta_i(w) = \|w\| - \|z\| \leq 0$, we get $\|u\| \leq 1$, and the proof for v is symmetric.

It remains to prove $u \in U_0$ and $v \in V_0$. Let $f \in F$. The bad piece D_1 contributes only a norm-small term, because $\int_{D_1} (r + s + q) d\mu$ and $\|w_1\|$ are small. On a good piece D_i , $i \geq 2$, Lemma 1 gives

$$\|(\alpha_i/\gamma_i)z_i - x_i\|_1 \leq 2\delta\alpha_i.$$

Therefore

$$u_i - x_i = (\alpha_i/\gamma_i)(w_i - z_i) + ((\alpha_i/\gamma_i)z_i - x_i) + (\nu\kappa/\rho)w_i.$$

The first term is small under f by the definition of W , since $\alpha_i/\gamma_i \leq 1/\lambda$. The second term is norm-small by the directional averaging lemma. The third term has total norm at most $|\kappa|$, which is small. Summing over the finitely many pieces gives

$$|f(u - x)| < \eta.$$

Thus $u \in U_0$. The same argument, using the second estimate in Lemma 1, gives $v \in V_0$. Hence $w = \lambda u + \nu v \in \lambda U + \nu V$.

Since every $w \in W$ belongs to $\lambda U + \nu V$, z is a relative weak interior point of $\lambda U + \nu V$. This proves *CWO-S* for $L_1(\mu, X)$. $\square$

Remark 1. *The proof deliberately avoids identifying $L_1(\mu, X)^*$ with $L_\infty(\mu, X^*)$. Instead, it localizes the finitely many functionals that define the weak neighbourhoods by composing them with the bounded coordinate multipliers $f \mapsto \mathbf{1}_D f$. Thus no Radon–Nikodym property of X^* is used.*

Verification Notes

The proof is non-computational. The points for review are:

- the strict-convexity step forcing pointwise positive collinearity;
- the finite partition of the direction map $e = z/\|z\|$;
- the use of localized functionals $f \circ (\mathbf{1}_D \cdot)$ instead of an $L_\infty(\mu, X^*)$ -representation;
- the ℓ_1 -norm bookkeeping copied from the source proof.

Limitations

This is still a partial result. It proves the $L_1(\mu, X)$ theorem for the large natural class of strictly convex target spaces, but it does not cover all *CWO-S* spaces. In non-strictly-convex *CWO-S* spaces, equality in the pointwise triangle inequality no longer forces a common direction, and the argument above no longer supplies norm-small local errors. The injective tensor-product question remains open here as well.

Novelty and Literature Check

I searched the run indexes and local corpus for arXiv:2503.09182, *CWO-S*, $L_1(\mu, X)$, strictly convex targets, and injective tensor variants. The existing run packet for this paper proves the arbitrary ℓ_1 -sum/atomic case, and the older arXiv:1806.10693 packet treats a separate finite-dimensional tensor subcase. Neither contains the strictly convex $L_1(\mu, X)$ argument.

Bounded web searches for “ $L_1(\mu, X)$ strictly convex CWO-S”, “strictly convex property CWO-S Bochner”, and close variants found the source paper and nearby CWO literature, but no exact statement of the theorem above. Novelty confidence is moderate pending expert review.

Human Review Recommendation

Review the proof as a claimed strict-convex upgrade of the source theorem, not as a full solution of the X has $CWO-S$ question. The most important line to audit is the final estimate of $f(u - x)$: it combines localized weak control of $\mathbf{1}_{D_i}(w - z)$, norm-small fixed directional errors, and the small global correction κ .

References

- [1] R. Haller, P. Kuuseok, and M. Poldvere, *Norm-one points in convex combinations of relatively weakly open subsets of the unit ball in the spaces $L_1(\mu, X)$* , arXiv:2503.09182, 2025. <https://arxiv.org/abs/2503.09182>
- [2] T. A. Abrahamsen, J. Becerra Guerrero, R. Haller, V. Lima, and M. Poldvere, *Banach spaces where convex combinations of relatively weakly open subsets of the unit ball are relatively weakly open*, *Studia Math.* **250** (2020), no. 3, 297–320. <https://arxiv.org/abs/1806.10693>